BSc 2023/2024

For internal students of Imperial College.

Mathematics Foundations (BUSI40001)

2 hours

XXXday ABth January 2024, 09:00 – 11:00

Instructions for the exam

- For Section A, internet access with ChatGPT is allowed and a computer with R or Python is needed.
- Section B is closed-book and paper-based. Section B is to made accessible to the students after completing Section A.
- Answer ALL questions (Please note that not all questions carry equal marks, the mark breakdown is indicated throughout the exam).
- It is important to write neatly and clearly.
- If a question or part of a question is unclear to you, state any assumptions you need in order to complete the question.

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SECTION A

You have 45 minutes for Section A. Answer all questions. You must close your laptop after 45 minutes.

1. (19 marks) The blood pressures, p mmHg, and the ages, t years, of 7 hospital patients are shown in the table below.

Patient	A	B	C	D	E	F	G
t	42	74	48	35	56	26	60
p	98	130	120	88	182	80	135

This question is about estimating blood pressure given age.

(a) Write out the linear regression in the matrix form

$$\underline{y} = \mathbf{X}\underline{\beta} + \underline{\epsilon},$$

where $\underline{y}$, $\mathbf{X}$, $\underline{\beta}$, and $\underline{\epsilon}$ are to be defined. [2 marks]

[1/2 a mark for each vector]

$$\underline{y} = (98, 130, 120, 88, 182, 80, 135)^\top$$

is the 7×1 vector of blood pressure measurements,

$$\underline{x} = (42, 74, 48, 35, 56, 26, 60)^\top$$

is the 7×1 vector of ages, $\underline{e} = (1, \ldots, 1)^\top$ is 7×1 vector of ones.

$$\mathbf{X} = [\underline{e}, \underline{x}]$$

is the 7 by 2 design matrix,

$$\underline{\beta} = [\beta_1, \beta_2]^\top$$

is the 1 by 2 parameter vector, and $\underline{\epsilon}$ is 7×1 error vector.

(b) Show that the expression for $\underline{\beta}$ which minimizes $\underline{\epsilon}^\top \underline{\epsilon}$ is given by

$$\widehat{\underline{\beta}} = \left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\underline{y}.$$

[5 marks]

Observe that $\underline{\epsilon}^\top\underline{\epsilon}$, where $\underline{\epsilon} = \underline{y} - \mathbf{X}\underline{\beta}$, is minimized by projecting $\underline{y}$ orthogonally into the column space of $\mathbf{X}$. Therefore, we want to choose $\underline{\beta}$ such that $\mathbf{X}^\top(\underline{y} - \mathbf{X}\underline{\beta}) = \underline{0}$. We shall denote the parameter vector which satisfies the latter equation via $\widehat{\underline{\beta}}$ and so

$$\mathbf{X}^\top(\underline{y} - \mathbf{X}\widehat{\underline{\beta}}) = \underline{0}$$

Therefore,

$$\mathbf{X}^\top\underline{y} = \mathbf{X}^\top\mathbf{X}\widehat{\underline{\beta}}.$$

Now $\mathbf{X}^\top\mathbf{X}$ is a 2 by 2 square matrix. Now

$$\widehat{\underline{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\underline{y}.$$

(c) Show that the fitted values are given by

$$\underline{\hat{y}} = \mathbf{H}\underline{y}$$

where

$$\mathbf{H} \equiv \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T$$

[3 marks]

The fitted values of the dependent variable are given by $\underline{\hat{y}} = \underline{\widehat{\beta}}\mathbf{X}$. Therefore,

$$\underline{\hat{y}} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\underline{y},$$

and so

$$\underline{\hat{y}} = \mathbf{H}\underline{y},$$

where

$$\mathbf{H} \equiv \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T.$$

(d) Show that $\mathbf{H}$ is idempotent and symmetric. [4 marks]

$$\begin{aligned}
\mathbf{H}^2 &= \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T \\
&= \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T \\
&= \mathbf{H},
\end{aligned}$$

and so $\mathbf{H}$ is idempotent. Now,

$$\begin{aligned}
\mathbf{H}^\top &= [\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T]^\top \\
&= [\mathbf{X}^\top]^\top[\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}]^\top \\
&= \mathbf{X}[\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}]^\top \\
&= \mathbf{X}[\left(\mathbf{X}^T\mathbf{X}\right)^{-1}]^\top\mathbf{X}^\top \\
&= \mathbf{X}[\left(\mathbf{X}^T\mathbf{X}\right)^\top]^{-1}\mathbf{X}^\top \\
&= \mathbf{X}\left[\left(\mathbf{X}^\top(\mathbf{X}^T)^\top\right)\right]^{-1}\mathbf{X}^\top \\
&= \mathbf{X}\left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1}\mathbf{X}^\top \\
&= \mathbf{H},
\end{aligned}$$

and so $\mathbf{H}$ is symmetric.

(e) Estimate the blood pressure of an 80 year old. You should do this using either R or Python [5 marks]

You need a calculator or a computer!

$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} 7 & 341 \\ 341 & 18181 \end{pmatrix}$$

$$\left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1} = \begin{pmatrix} \frac{18181}{10986} & -\frac{341}{10986} \\ -\frac{341}{10986} & \frac{7}{10986} \end{pmatrix}$$

$$\left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1}\mathbf{X}^\top = \begin{pmatrix} \frac{3859}{10986} & -\frac{2351}{3662} & \frac{1813}{10986} & \frac{1041}{1831} & -\frac{305}{3662} & \frac{3105}{3662} & -\frac{2279}{10986} \\ -\frac{47}{10986} & \frac{59}{3662} & -\frac{5}{10986} & -\frac{16}{1831} & \frac{17}{3662} & -\frac{53}{3662} & \frac{79}{10986} \end{pmatrix}$$

$$\begin{aligned} \underline{\widehat{\beta}} &= \left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1}\mathbf{X}^\top\underline{y} \\ &= \begin{pmatrix} \frac{499505}{10986} \\ \frac{16583}{10986} \end{pmatrix} \\ &= \begin{pmatrix} 45.4674 \\ 1.50947 \end{pmatrix} \end{aligned}$$

$$\beta_1 + \beta_2 80 = 166.225$$

Estimate blood pressure is 166.225 mmHg.

2. (20 marks) Find and classify all the critical points of the function $f(x, y) = (x + y)(xy + xy^2)$, where $x, y \in \mathbb{R}$. You may find it useful to use R or Python to draw plots and perform calculations.

This question has a sting in its tail.

It's easy to find all the critical points. All one needs to do is solve the equations $f_x(x, y) = 0$ and $f_y(x, y) = 0$, which reduce to

$$xy^2 + \left(y^2 + y\right)(x + y) + xy = 0,\ xy^2 + xy + (x + y)(2xy + x) = 0$$

The solutions are given by the coordinates in the set

$$\{(0, 0),\ (0, -1), (1, -1), (3/8, -3/4)\}.$$

All you need is to write sensible code for this, which should be easy.

To classify the critical points, one must evaluate the Hessian at each critical point and computes its eigenvalues, which is easy using code.

At $(0, 0)$ the Hessian is the zero matrix, so we need to look at third derivatives, which is the hard part of the question.

At $(0, -1)$ the Hessian has eigenvalues of differing sign, so this is a saddle point.

At $(1, -1)$ the Hessian has eigenvalues of differing sign, so this is also a saddle point.

At $(3/8, -3/4)$ the Hessian has two strictly negative eigenvalues and so this is a max.

This is enough for 15 marks.

For the final 5 marks, the third derivatives at $(0, 0)$ need to be analyzed correctly. Ex-

panding $f(x, y)$ to third order around $(0, 0)$ gives

$$\begin{aligned}
f(x,y) &= f(0,0) + f_x(0,0) + f_y(0,0) + (x,y)\mathbf{H}_f(0,0)(x,y)^\top \\
&\quad + \frac{1}{3!}\left[x^3 f_{xxx}(0,0) + 3x^2 y f_{xxy}(0,0) + 3xy^2 f_{xyy}(0,0) + y^3 f_{yyy}(0,0)\right] + \dots \\
&= \frac{1}{3!}\left[3x^2 y f_{xxy}(0,0) + 3xy^2 f_{xyy}(0,0) +\right] + \dots \\
&= \frac{1}{3!}\left[3x^2 y \times 2 + 3xy^2 \times 2\right] + \dots \\
&= x^2 y + xy^2 + \dots
\end{aligned}$$

We can see that $x^2y + xy^2$ can be positive or negative, so this is a saddle point.

SECTION B

You have 1 hour 15 minutes for Section B. Answer all questions.

1. (30 marks) Consider a real sequence $(a_n)_n$

(a) Define what is meant by a subsequence of $(a_n)_n$. [3 marks]

Let (a_n) be a sequence of real numbers, and let $n_1 < n_2 < n_3 < n_4 < n_5 < \ldots$ be a strictly increasing sequence of natural numbers. Then the sequence

$$(a_{n_1}, a_{n_2}, a_{n_3}, a_{n_4}, a_{n_5}, \ldots)$$

is called a subsequence of (a_n) and is denoted by (a_{n_k}), where $k \in \mathbb{N}$ indexes the subsequence.

(b) Prove that a sequence converges to a limit if and only if every subsequence converges to that same limit, i.e.

$$a_n \to a \iff (\text{for every subsequence } a_{n_k}) a_{n_k} \to a$$

[5 marks]

We start by assuming $\lim a_n = a$, i.e.

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)|a_n - a| < \epsilon.$$

Now, for every $m : \mathbb{N} \to \mathbb{N}$ such that $m(n+1) \geq m(n)$, we can find an M such that $m(M) > N$. Therefore,

$$(\forall \epsilon > 0)(\exists M \in \mathbb{N})(\forall n \geq M)|a(m(n)) - a| < \epsilon.$$

Hence, $a_{m_n} \to a$.
Now suppose that for every $m : \mathbb{N} \to \mathbb{N}$ such that $m(n+1) \geq m(n)$,

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)|a(m(n)) - a| < \epsilon.$$

If this is true for every $m : \mathbb{N} \to \mathbb{N}$ such that $m(n+1) \geq m(n)$, we can just choose $m(n) = n$, so that

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)|a_n - a| < \epsilon.$$

(c) What is meant by the peak of a sequence $(a_n)_n$? [1 mark]

a_m is a peak of the sequence (a_n) if $(\forall n > m) a_m \geq a_n$.

(d) Prove that every sequence contains a monotone subsequence. [8 marks]

A sequence will either have a fixed number of peaks, say $N \in \mathbb{N}$ or the number of peaks, like the natural numbers, will never end.
If the number of peaks never ends, let a_{n_k} be the k'th peak. Then for every $k \in \mathbb{N}$, we have $a_{n_k} \geq a_{n_{k+1}}$. Hence, (a_{n_k}) is a monotone decreasing subsequence.
Now, we consider the case where there are fixed numbers of peaks. Without loss of generality, ket a_K be the final peak. Now, $(\forall n \geq K) a_n \leq a_K$. Define $a_{n_1} = a_{K+1}$ and let a_{n_2} the the next term in the sequence which is greater than or equal to a_{n_1}. Such a term must exist, because a_{n_1} is not a peak. Similarly, let a_{n_3} the the next term in the sequence which is greater than or equal to a_{n_2}, which must exist, because a_{n_2} is

not a peak. Continuing in this way, we construct a monotone increasing subsequence.

(e) State the Monotone Convergence Theorem without proof. [1 mark]

If a sequence is monotone and bounded, then it converges.

(f) State and prove the Bolzano-Weierstrass Theorem. You may use results you have proved or stated earlier in this question. [3 marks]

Bolzano-Weierstrass Theorem

Every bounded sequence contains a convergent subsequence.

We have show that every sequence contains a monotone subsequence. Therefore, every bounded sequence contains a bounded and monotone subsequence, which by the Monotone Convergence Theorem, must converge.

(g) Define

$$a_n = \left(\frac{n^2-1}{n^2+1}\right)\cos\left(\frac{2n\pi}{3}\right).$$

Decide whether or not the sequence $(a_n)_n$ converges and prove your conjecture is correct. You may use the Algebraic Limit Theorem without proof. [9 marks]

Using the Algebraic Limit Theorem,

$$\lim a_n = \lim\left(\frac{n^2-1}{n^2+1}\right) \times \lim\cos\left(\frac{2n\pi}{3}\right)$$

We first consider the sequence (b_n), where $b_n = \frac{n^2-1}{n^2+1}$, which we can rewrite as

$$b_n = \frac{1-n^{-2}}{1+n^{-2}}$$

Again, using the Algebraic Limit Theorem, we have

$$\lim b_n = \frac{\lim 1-n^{-2}}{\lim 1+n^{-2}}.$$

A further application of the the Algebraic Limit Theorem gives $\lim 1-n^{-2} = 1-\lim n^{-2}$ and $\lim 1+n^{-2} = 1+\lim n^{-2}$. We now prove that $\lim n^{-2} = 0$.

Recall that, by definition $\lim a_n = a$ iff

$$(\forall\epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)|a_n - a| < \epsilon.$$

Now note that

$$\left|n^{-2}\right| = n^{-2}.$$

Therefore,

$$\left|n^{-2}\right| < \epsilon \iff n^{-2} < \epsilon \iff n > \frac{1}{\sqrt{\epsilon}}.$$

By the Archimedean property,

$$(\forall\epsilon > 0)(\exists N \in \mathbb{N})N > \frac{1}{\sqrt{\epsilon}}.$$

The above statement is equivalent to

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N})\left|N^{-2}\right| < \epsilon.$$

Hence,

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)\left|n^{-2}\right| < \epsilon.$$

Therefore, $\lim n^{-2} = 0$.
It follows that

$$\lim b_n = 1.$$

Hence,

$$\lim a_n = \lim \cos\left(\frac{2n\pi}{3}\right).$$

We shall now prove that $\left(\cos\left(\frac{2n\pi}{3}\right)\right)_n$ does not converge.
We shall assume $(\exists L \in \mathbb{R})(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)\left|\cos\left(\frac{2n\pi}{3}\right) - L\right| < \epsilon$. By the Archimedean property we can choose $n = 3k \geq N$, and so

$$|\cos(2k\pi) - L| < \epsilon \iff |1 - L| < \epsilon \iff 1 - L < \epsilon \wedge L - 1 < \epsilon \iff 1 - \epsilon < L < 1 + \epsilon.$$

By the Archimedean property we can also choose $n = 3k + 1 \geq N$, and so $\cos\left(\frac{2n\pi}{3}\right) = \cos\left(\frac{2(3k+1)\pi}{3}\right) = -\frac{1}{2}$. Hence,

$$\left|-\frac{1}{2} - L\right| < \epsilon \iff -\frac{1}{2} - L < \epsilon \wedge L + \frac{1}{2} < \epsilon \iff -\frac{1}{2} - \epsilon < L < -\frac{1}{2} + \epsilon.$$

Therefore,

$$(\exists L \in \mathbb{R})(\lim \cos\left(\frac{2n\pi}{3}\right) = L) \Rightarrow (\forall \epsilon > 0)(1 - \epsilon < L < 1 + \epsilon \wedge -\frac{1}{2} - \epsilon < L < -\frac{1}{2} + \epsilon)$$

When $\epsilon = 1/2$, the inequalities cannot all hold, so we have a contradiction. Therefore, $\left(\cos\left(\frac{2n\pi}{3}\right)\right)_n$ does not converge. Therefore $(a_n)_n$ does not converge.

2. (24 marks) Let V be a vector space over $\mathbb{R}$.

(a) Suppose V is finite dimensional. Let $\mathcal{S}$ be a finite subset of V. What does it mean for $\mathcal{S}$ to be

i. a linearly independent set [2 marks]

The vectors in a set $\mathcal{S} = \{x_1, \ldots, x_n\}$ are said to be **linearly independent** if

$$\sum_{i=1}^{n} \lambda_i x_i = 0 \iff (\forall i \in \{1, \ldots, n\})\lambda_i = 0.$$

ii. a spanning set for V [2 marks]

A set of vectors $\mathcal{S} = \{v_1, \ldots, v_n\}$ in a vector space V over $\mathbb{R}$ is said to be a **spanning set** of V if every vector in V can be expressed as a linear combination of the vectors

in $\mathcal{S}$. In other words, $\mathcal{S}$ spans V (or $\mathcal{S}$ is a spanning set of V)

$$(\forall v \in V)(\exists a_1, a_2, \ldots, a_n \in \mathbb{R})v = \sum_{i=1}^{n} a_i v_i.$$

iii. a basis for V? [2 marks]

A set of vectors $\mathcal{S} = \{v_1, \ldots, v_n\}$ in a vector space V over $\mathbb{R}$ is said to be a **basis** of V it is a linearly independent spanning set of V.

(b) What does it mean for $W \subseteq V$ to be a subspace? For the real vector space $V = \{f : \mathbb{R} \to \mathbb{R}\}$ with subset $W = \{f \in V : f'' + f = 0\}$ determine whether W is a subspace of V or not. Justify your answer. If W is a subspace of V, give, without justification, a basis for W. [5 marks]

If W is a vector space over $\mathbb{R}$ with the same operations as V, then W is a **subspace** of V. [1 mark]

We consider the ordinary differential equation $f''(x) + f(x) = 0$, where $x \in \mathbb{R}$. The general solution is of the form

$$f(x) = A\cos x + B\sin x$$

where $A, B \in \mathbb{R}$. Therefore, $W = \{A\cos x + B\sin x : A, B \in \mathbb{R}\}$, which is the set of linear combinations of the functions $\sin x$ and $\cos x$. We can now see that

i. The zero vector, 0, is in W.
ii. $(\forall u, v \in W)$, $u + v$ are in W
iii. $(\forall a \in \mathbb{F})(\forall v \in W)a \cdot v \in W$

Therefore, W is a subspace of V. [3 marks]

Students may instead show that directly that $f = 0$ satisfies $f'' + f = 0$, and so $0 \in W$. Also, $u, v \in W \iff u'' + u = 0 \wedge v'' + v = 0 \Rightarrow (u + u)'' + u + v = (u'' + u) + (v'' + v) = 0 + 0 = 0 \iff u + v \in W$. And, $(\forall a \in \mathbb{F})(\forall v \in W)a \cdot v \in W$, because $(\forall a \in \mathbb{R})(v'' + v = 0) \Rightarrow (\forall a \in \mathbb{R})((av)'' + av = a(v'' + v) = 0)$.

The basis of W is $\{\sin x, \cos x\}$. [1 mark]

(c) What is meant by an inner product $\langle \cdot, \cdot \rangle$ on V? Let U be a subspace of V. Show that

$$U^{\perp} = \{v \in V : (\forall u \in U)\langle v, u\rangle = 0\}$$

is a subspace of V. [4 marks]

An **inner product** is a map $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$, which satisfies the following 3 axioms $(\forall x, y, z \in V)(\forall a \in \mathbb{R})$:

i. $\langle x, x\rangle \geq 0$ with equality iff $x = 0$ (positive definiteness)
ii. $\langle ax, y\rangle = a\langle x, y\rangle$ (homogeneity in the 1st argument)
iii. $\langle x + y, z\rangle = \langle x, z\rangle + \langle y, z\rangle$ (additivity in the 1st argument)

[1 mark]

$U^{\perp}$ is clearly a subset of V

$\langle 0, u\rangle = 0 \Rightarrow 0 \in U^{\perp}$

$$\begin{aligned} &v_1, v_2 \in U^{\perp} \Rightarrow \langle v_1, u\rangle = 0 \wedge \langle v_2, u\rangle = 0 \\ &\Rightarrow \langle v_1 + v_2, u\rangle = 0 (\text{exploiting additivity in the 1st argument}) \\ &\Rightarrow v_1 + v_2 \in U^{\perp} \end{aligned}$$

Via homogeneity in the 1st argument, we have $(\forall a \in \mathbb{R})(\langle v_1, u\rangle = 0 \Rightarrow \langle av_1, u\rangle = 0)$. Hence, $(\forall a \in \mathbb{F})(\forall v \in U^\perp) a \cdot v \in U^\perp$ Therefore, $U^\perp$ is a subspace of V.
[3 marks]

(d) Let $Z = C[-1,1]$ denote the vector space of continuous real-valued functions on $[-1,1]$ with inner product

$$\langle f, g\rangle_Z = \int_{-1}^{1} f(x)g(x)\mathrm{d}x.$$

[You are not required to show that $\langle \cdot, \cdot \rangle_Z$ is an inner product and you do not need to know what a continuous function is.] Let

$$X = \{f \in Z : (\forall x \in \mathbb{R}) f(-x) = f(x)\}; \quad Y = \{f \in Z : (\forall x \in \mathbb{R}) f(-x) = -f(x)\}.$$

Show that $Z = X \oplus Y$. [9 marks]

Let $Z = C[-1,1]$ be the vector space of continuous real-valued functions on $[-1,1]$ with the inner product

$$\langle f, g\rangle_Z = \int_{-1}^{1} f(x)g(x)\,\mathrm{d}x.$$

Define

$$X = \{f \in Z : f(-x) = f(x) \text{ for all } x\}, \quad Y = \{f \in Z : f(-x) = -f(x) \text{ for all } x\}.$$

Then $Z = X \oplus Y$.

To show $Z = X \oplus Y$, we must demonstrate two things:

1. Every function in Z can be expressed as the sum of a function in X and a function in Y.

Let $f \in Z$. Define the functions $f_e, f_o : [-1,1] \to \mathbb{R}$ by

$$f_e(x) = \frac{f(x) + f(-x)}{2}, \quad f_o(x) = \frac{f(x) - f(-x)}{2}.$$

Then f_e is an even function (in X) and f_o is an odd function (in Y), and we have

$$f(x) = f_e(x) + f_o(x) \text{ for all } x \in [-1,1].$$

2. X and Y are orthogonal subspaces.

Let $f \in X$ and $g \in Y$. Then

$$\langle f, g\rangle_Z = \int_{-1}^{1} f(x)g(x)\,\mathrm{d}x.$$

Since f is even and g is odd, $f(x)g(x)$ is odd. Thus, the integral of an odd function over a symmetric interval is zero, and so $\langle f, g\rangle_Z = 0$.

Therefore, $Z = X \oplus Y$.

End of Paper